ME 144L Post Lab 4

Alejandro Diaz
TA: Gabrielle Naquila

February 24, 2026

1 Sensor Calibration

The potentiometer sensor was calibrated by recording voltage readings at five known angular positions of the pendulum. The pendulum was then manually placed at each angle, and verified with a handheld angle measurement device until angular displacement fell within the expected 45 deg intervals. The corresponding voltage from the Arduino DAQ serial monitor was recorded. A linear regression was then applied to relate voltage to angular displacement.

1.1 Calibration Data

Voltage Reading (V)	**Angular Displacement θ (deg)**
0.43	-89.7
1.31	-45.3
2.46	0.3
3.54	45.4
4.26	89.9

Table 1: Raw Calibration Data of Voltage Readings and Corresponding Measured Angles

1.2 Calibration Results

Applying linear regression from the pot_sensor_calibration_example.py file to the calibration data provided the following linear model, which takes measured voltage as a function of angular displacement of the pendulum.

$$\theta = K \cdot V + \theta_0 \tag{1}$$

where K is the calibration coefficient, or the slope of the calibration plot, and θ_0 is the intercept variable. The regression calibration produced the following values and their respective units:

$$K = 45.269\ \frac{\text{deg}}{\text{V}}, \qquad \theta_0 = -108.525\ \text{deg}, \qquad R^2 = 0.9954$$

The R^2 value of 0.9954 suggests a strong linear correlation in data that provides insight that this calibration process using handheld angular measurement devices is consistent. After further testing in simulation, an additional manual offset correction of -8.72 deg was applied to the intercept after comparing the calibration model's predicted angle to a known resting position observed in the serial log. This brought the true effective intercept to:

$$\theta_{0,\text{corrected}} = -108.525 - 8.72 = -117.245\ \text{deg}$$

The strong linear relationship between voltage output and angular displacement validates the use of a linear model for sensor calibration. Below is the calibration plot showing the raw data across the 5 measured points and the linear interpolation function through them.

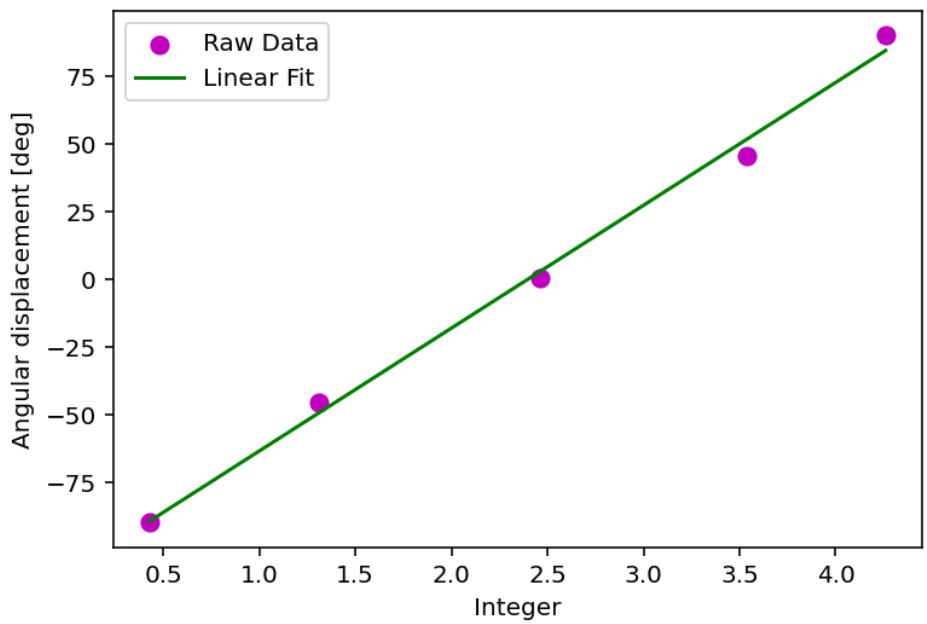

Figure 1: Potentiometer Calibration Data and Linear Regression Fit

2 Compound Pendulum Simulation vs. Measured Response

The compound pendulum simulation was tuned to match the measured angular response by adjusting the linear viscous damping coefficient b and the Coulomb friction torque coefficient τ_0. The equation of motion governing the compound pendulum is:

$$\dot{\omega} = \frac{\tau_{net}}{J_O} \tag{2}$$

Where τ_{net} represents the sum of moments in the system, which can be expanded into three terms: linear viscous damping, coulombic friction, and torque induced by gravity.

$$J_O\dot{\omega} = -b\omega - \tau_0 \operatorname{sgn}(\omega) - M_{tot}\, gL_c \sin(\theta) \tag{3}$$

2.1 Model Parameters — Pendulum 3

where J_O is the mass moment of inertia about the pivot, L_c is the distance from the pivot to the center of gravity, and M_{tot} is the total pendulum mass. For the physical system used, we can model this as pendulum 3 from the previous pre-lab conducted. The physical parameters of this system were determined using moment of inertia and volume calculations under the assumption of the expected density for 6061 aluminum of $\rho = 2700\ \mathrm{kg/m^3}$.

Parameter	Symbol	Value
Total mass	M_{tot}	0.1574 kg
Distance to center of gravity (from pivot)	L_c	0.1992 m
Mass moment of inertia about pivot	J_O	6.479×10^{-3} kg·m^2
Gravitational acceleration	g	9.81 m/s^2
Initial Linear damping coefficient	b	4.375×10^{-6} N·m·s/rad
Initial Coulomb friction torque coefficient	τ_0	1.875×10^{-3} N·m

Table 2: Compound pendulum model parameters for Pendulum 3.

2.2 Initial Conditions

Condition	Symbol	Value
Initial angle	θ_0	$-90\,\text{deg}$
Initial velocity	$\dot{\theta}_0$	0 rad/s
Start time	t_0	3 s
End time	t_f	30 s

Table 3: Initial conditions for the simulation.

The simulation was integrated using a fixed-step fourth-order Runge-Kutta (RK4) method with 501 time steps over the interval from 3 to 30 seconds. This 3-second offset allows for later tuning when comparing potentiometer data to simulated conditions for an ideally identical pendulum case.
The estimated moment of inertia could be adjusted to align with real, recorded data, as this would be the critical controlled dimension that would change the natural frequency of the pendulum.

After manual tuning, we can gain insight into the impacts of the coulombic friction and linear damping through trial and error of aligning the simulation to real data. Below is a table containing the tuned values and the multiplication factor applied to modify the initial simulation parameters for alignment.

Parameter	Symbol	Value	Factor
Mass moment of inertia about pivot	J_O	7.126×10^{-3} kg·m^2	1.1
Linear damping coefficient	b	4.375×10^{-6} N·m·s/rad	1
Coulomb friction torque coefficient	τ_0	5.062×10^{-3} N·m	2.7

Table 4: Compound Pendulum Tuned Parameters for Simulation

2.3 Simulated vs. Measured Results

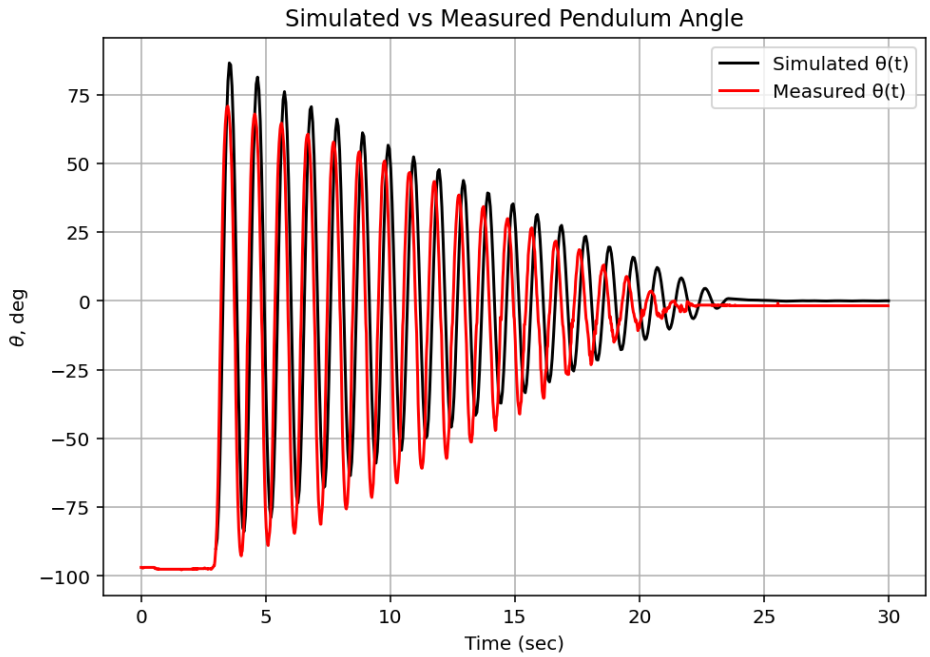

Figure 2: Simulated (black) and Measured (red) Pendulum Angle over Time

The simulation closely tracks the measured decay envelope and oscillation frequency, confirming that the selected values of b and τ_0 provide a good model fit to the physical system.

3 Dominant Form of Friction

To determine which friction mechanism dominates the pendulum's energy dissipation, the total friction torque $\tau_{friction}$, the viscous damping contribution $\tau_b = -b\omega$, and the Coulomb friction contribution $\tau_C = -\tau_0 \operatorname{sgn}(\omega)$ were extracted from the simulation at each time step and plotted together.

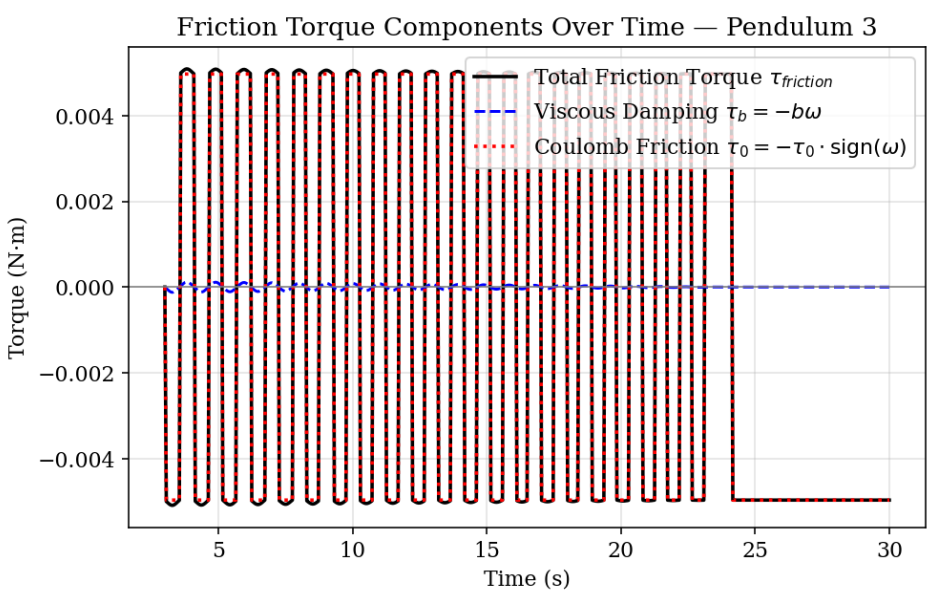

Figure 3: Total Friction, Viscous Damping, and Coulomb Friction Torques Over Time in Simulation

The magnitudes of each friction component, computed over the simulation duration, are summarized below:

Torque Component	**Mean Magnitude (N·m)**	**Peak Magnitude (N·m)**
Viscous damping $\lvert\tau_b\rvert$	3.0×10^{-5}	1.2×10^{-4}
Coulomb friction $\lvert\tau_C\rvert$	4.96×10^{-3}	4.97×10^{-3}

Table 5: Comparison of Friction Torque Component Magnitudes

3.1 Conclusion

The data clearly shows that Coulombic friction is the dominant form of friction in this pendulum system. The mean Coulomb friction torque (4.96×10^{-3} N·m) exceeds the mean viscous damping torque (3.0×10^{-5} N·m) by a factor of approximately 165. As shown in Figure 4, the Coulomb friction torque remains nearly constant in magnitude throughout the motion, changing sign only when the pendulum reverses direction. The viscous component, which is proportional to angular velocity ω, remains negligible in comparison at all times.

This is physically consistent with the pendulum's pivot, which exhibits significant dry contact friction rather than fluid-film damping. The result also explains the characteristic "staircase" amplitude decay observed in the measured data: Coulomb friction dissipates a fixed amount of energy per half-cycle regardless of amplitude.